CHIDU UMEH

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Professional Summary: M.S. Physics graduate (UCF, May 2026) specializing in adiabatic spintronics, with 4+ years of hands-on experimental research experience designing rigorous studies, interpreting complex datasets, and building reproducible analytical pipelines in Python and MATLAB. Completed the Erdos Institute Quant Finance Bootcamp, building and backtesting a Black-Scholes-Merton implied volatility model that won Top Project out of 52 submissions, and regularly applies that same quantitative rigor to physics research. Ships production software independently, including a full-stack fintech application live at splittank.com and an AI-assisted analytics platform processing 300,000+ records deployed across all 50 states. Daily user of Claude and Claude Code as core research and development tools, with a peer-reviewed publication and national conference presentation demonstrating the ability to execute full research cycles and communicate findings to technical and non-technical audiences alike. U.S. citizen eligible to obtain a security clearance.

SKILLS

Tools: Tableau, Jupyter Notebooks, LaTeX, Creo CAD, Microsoft Office Suite (Excel, PowerPoint), LLM-Assisted Development (Claude, Claude Code, ChatGPT Codex)
Specialties: Experimental Physics, Quantitative Finance & Options Modeling, Spintronics Research, Rigorous Evaluation Design, AI-Assisted Software Development, Full-Stack Web Development, Statistical Modeling, Peer-Reviewed Research
Languages: Python (Pandas, NumPy, Scikit-learn, TensorFlow), SQL (PostgreSQL), R, MATLAB, Fortran, Git

PROFESSIONAL EXPERIENCE

University of Central Florida, Orlando, FL — **Jan 2025 - Present**
Graduate Research Assistant
• Designed and executed rigorous experimental evaluations of electronic resonance behavior across 10+ antiferromagnetic and ferromagnetic material configurations, systematically identifying where theoretical models diverged from observed dynamics and building Python pipelines to close the gap.
• Integrated domain-specific experimental tools and software into reproducible research workflows, applying Claude and LLM-assisted development to accelerate pipeline development by an estimated 35% and expand test coverage across 5+ active research configurations.
• Designed custom experimental hardware and measurement protocols for a 5-person research team, iterating on 3 generations of experimental setups and reducing measurement uncertainty by an estimated 20% through systematic protocol refinement.

Split Tank — **Jan 2026 - May 2026**
• Built and deployed a full-stack web application integrating live EIA gas price feeds, FuelEconomy.gov MPG lookups, and GPS-based location detection to calculate real-time road trip cost splits.
• Engineered payment deep links for Venmo, Cash App, Zelle, and Apple Pay with pre-filled split amounts, delivering a production-ready fintech tool serving live users at splittank.com.
• Owned the product end to end as a solo builder, from API integration and backend logic through UI design and deployment, without a team or existing codebase to build on.

Independent Project — **Jan 2026 - Apr 2026**
U.S. Gun Violence Intelligence Tool
• Architected an end-to-end scientific data pipeline processing 300,000+ CDC records spanning 25 years, using Claude and ChatGPT Codex as core development partners to integrate ingestion, statistical modeling, and interactive visualization into a single platform.
• Designed and shipped a production-ready AI-assisted analytics tool as a solo researcher, deploying interactive, state-by-state visualizations covering all 50 states.
• Demonstrated high-agency execution and comfort with fast-paced development cycles, taking the project from raw federal data to a public, shareable platform within a defined timeframe.

Erdos Institute (selective STEM graduate fellowship, 52-person admitted cohort) — **Sep 2025 - Nov 2025**
Quant Finance Bootcamp Participant

• Awarded Top Project out of 52 submissions for designing and executing a rigorous quantitative evaluation framework from scratch, systematically identifying model performance gaps across 250+ trading days.
• Built a Black-Scholes-Merton implied volatility model and backtested it against realized volatility on the VanEck Africa ETF (AFK), surfacing a systematic risk premium (IV > RV, correlation 0.27) that showed where the model was confidently wrong.
• Delivered production-ready code and structured findings to a panel of 10 senior evaluators, competing against 51 other submissions from a selective, cohort-based STEM fellowship.

University of Central Florida, Orlando, FL **Jan 2024 - Jan 2025**
Graduate Research Assistant
• Executed 50+ first-principles DFT simulations, modifying Fortran scripts to probe electronic structure, spin states, and phonon behavior across 3 defect configurations.
• Identified systematic discrepancies between DFT-predicted phonon behavior and experimental observations across 3 defect configurations, documenting failure modes and refining computational methodologies to improve model accuracy by an estimated 15%.

Autura **Summer 2023**
Data Analyst Intern
• Queried and normalized auction and logistics data across multiple source systems using SQL, producing analytics-ready datasets that powered executive dashboards and informed regional strategy.
• Built Tableau dashboards joining U.S. Census demographics with internal logistics records, surfacing regional cost patterns and growth signals that directly shaped market expansion decisions.

Bates College Astronomy **Mar 2019 - May 2022**
Research Data Analyst
• Engineered reproducible Python research pipelines processing 50,000+ astrophysical observations, applying multivariate regression to identify systematic gaps between theoretical stellar models and observed spectral data.
• Contributed findings to a peer-reviewed publication cited across 3 research institutions, demonstrating the ability to carry a research question from raw data to a published result.
• Identified and resolved systematic data quality issues across 4 photometric survey datasets, developing automated validation scripts that flagged anomalous observations and improved pipeline output reliability by an estimated 25%.

EDUCATION

University of Central Florida, Orlando, FL **2026**
M.S. Physics, Concentration: Spintronics (Materials Science & Electrical Engineering)

Bates College, Lewiston, ME **2022**
B.S. Physics, Minors: Computer Science & Applied Mathematics

CERTIFICATIONS

Certificate of Completion: AI Fluency Framework **Jul 2026**
Anthropic · Credential ID: qv3wudb37zt5

VOLUNTEERING

National Head Start Association **Jan 2024 - May 2026**
Guest Reader
• Lead regular bilingual (English/Spanish) reading sessions with preschool-age children at Southwood Head Start, building language exposure and foundational literacy skills.
• Collaborate with kitchen staff on meal preparation and nutrition education, and design and implement community garden improvements for outdoor environmental education.
• Received a third Certificate of Appreciation from Southwood Head Start for sustained bilingual literacy programming, garden development, and meal preparation support.